# Bulk Storage of Hydrogen

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## Abstract

The technical aspects and economics of bulk hydrogen storage in underground pipes, lined rock caverns (LRC) and salt caverns are analyzed. Hydrogen storage in underground pipes is more economical than in geological caverns for usable amounts <20-t-$H_2$. However, because the pipe material is a major cost factor, the capital and operating costs for this storage method do not decrease appreciably with an increase in the amount of stored $H_2$. Unlike underground pipes, the installed capital cost of salt caverns decreases appreciably from ~\$95/kg-$H_2$ at 100 t-$H_2$ stored to <\$19/kg-$H_2$ at 3,000 t-$H_2$ stored. Over the same scale, the annual storage cost decreases from ~\$17/kg-$H_2$ to ~\$3/kg-$H_2$. Like salt caverns, the installed capital cost of lined rock caverns decreases from ~\$160/kg-$H_2$ at 100 t-$H_2$ stored to <\$44/kg-$H_2$ at 3,000 t-$H_2$ stored. Storing >750-t usable $H_2$ requires multiple caverns. The cost of salt caverns scales more favorably with size because the salt caverns are larger than lined rock caverns and need to be added at a slower rate as the storage capacity is increased.

**Keywords:** *Bulk storage of hydrogen, underground hydrogen pipes, hydrogen storage in salt caverns, hydrogen storage in lined rock caverns*

## 1. Introduction

Massive deployment of hydrogen and fuel cells in use for power generation as well as transportation will necessarily require viable methods to store hydrogen at different time and length scales. Daily and seasonal storage of hydrogen can facilitate large-scale penetration of renewable energy by buffering the fluctuating imbalance between power/fuel production and

demand. In the industrial sector, bulk storage may be required to balance daily demands as well as to prevent interruption in supply when the plant undergoes maintenance.

At a small scale, hydrogen can be stored compactly as compressed gas at 350-950 bar in Type 2, 3 or 4 tanks. Storing gas at such high pressures requires the tanks to be reinforced with carbon fibers or pre-stressed concrete, which results in high capital and operating costs [1–4]. Typical applications of this method are hydrogen storage on board vehicles (5-10 kg at 350-700 bar in Type 3 or 4 tanks), buffer hydrogen storage at refueling stations (100 kg in Type 2 or 4 tanks), and tube trailers for hydrogen delivery to refueling stations (1,000 kg capacity, 540 bar, Type 4 tanks) [5–8].

At a medium scale, gaseous hydrogen could potentially be stored in spherical vessels or underground pipe facilities [3,9,10]. Spherical vessels have been utilized in Europe to store natural gas (NG) at a low pressure, ~20 bar, and thus require large spaces. Underground pipe facilities have been applied for seasonal storage of NG in Germany and Switzerland. A new pipe facility in Bocholt-Mussum (Germany) has the capacity to store 350,000 $m^3$ of natural gas at pressures up to 90 bar. The advantages of underground pipe storage facilities are the protection from adverse weather conditions and no requirement to have shaft structures, so that the land above ground can be used for other purposes such as farming.

At larger scales, hydrogen is stored cryogenically as a liquid at 20 K ($LH_2$) in spherical tanks [3,11–14]. The largest $LH_2$ storage facility at NASA's (National Aeronautics and Space Administration, U.S.A.) Kennedy Space Center has two 3,200-$m^3$ spheres, 21.3 m in diameter, each capable of storing 230 tonnes of $LH_2$ [13]. There are several issues with $LH_2$, however. Liquefaction of hydrogen is a costly and energy-demanding process [15–20]. State of the art liquefaction plants, based on a hydrogen compression/expansion sequence with a pre-cooling step,

consume 10-12 kWh/kg-$H_2$, which corresponds to one third of the lower heating value (LHV) of hydrogen and incurs ~$1.40/kg-$H_2$ levelized capital cost. That cost increases to ~$2.75/kg-$H_2$, if the cost of electricity is included [21]. Even though the cryogenic tanks are vacuum insulated, on a stored volume basis, the typical daily boil-off rate is 0.03% for the large spherical tank at the NASA complex, increasing to 1% for smaller cylindrical tanks [22,23]. Mitigation strategies exist to reduce boil-off by placing a liquid nitrogen shield between the insulation or using a closed-loop helium refrigeration system with an internal heat exchanger in the tank [24].

Underground geological storage is expected to be the most economic method for storage at substantial amounts of hydrogen [4,25–29]. It has been used widely, primarily in the natural gas and oil industry, for meeting seasonal and peak demands. The U.S. Energy Information Administration (EIA) estimates that there are 379 active underground fields for NG storage in the U.S., with ~3.8 trillion cubic feet of working gas capacity [30]. Eighty-eight percent of the operating storage is provided by depleted natural gas fields, 9% by aquifer reservoirs and 3% by salt caverns (although they account for 15% of the delivered capacity).

Conceptual studies have evaluated geological storage of gaseous hydrogen in depleted oil and gas fields, aquifers and salt caverns [31–37]. To date, hydrogen storage has been demonstrated in salt caverns [9], but not in depleted fields or aquifers, which require a caprock to seal hydrogen. The high diffusivity of hydrogen can result in losses in porous underground formations. In case of aquifers, hydrogen could potentially be lost by dissolving in the large contact area with the surrounding brine. This may be relevant, especially under circumstances of elevated temperature and excessive pressures [34,37]. Other issues with porous underground storage are the potential interactions between hydrogen and hydrocarbons dissolved in reservoir rocks through microbial metabolism [37–39]. In the presence of CO, $CO_2$ or sulfates, hydrogen can convert to $CH_4$ and

$H_2S$. Other potential issues are hydrogen reacting chemically (or electrochemically) with mineral deposits and initiating adverse changes in porosity and permeability [34]. Also, if hydrogen becomes contaminated, a separation step is required to provide fuel cell-quality hydrogen to the consumer.

Salt caverns are manmade artificial cavities with some unique properties of gas tightness and inertness. Depending on the depth of the cavity, storage pressures up to 25 MPa are feasible as are large capacities exceeding 1 million $m^3$. Pure hydrogen has been successfully stored without losses in salt caverns at three locations [9]: three caverns in Teesside, U.K. since 1972 (200,000 $m^3$), one in Clemens, TX since 1983 (580,000 $m^3$), and a more recent one in Moss, Texas since 2007 (566,000 $m^3$). Large salt formations are geographically limited, however. In the U.S., large salt bodies are primarily located in the Gulf Coast [35].

Where salt formations are not available, rock caverns can provide more possibilities close to hydrogen production facilities. Rock caverns have been constructed for storage of liquid hydrocarbons and natural gas and conceptually investigated for compressed air [40,41]. The caverns are constructed through typical mining operations. The concept of unlined rock caverns requires a competent rock with low permeability, free of fractures and fissures. Sufficient groundwater is needed to seal the rock or a water curtain must be applied in the surroundings for proper tightness [41]. However, the over-pressurized water system required around the caverns significantly increases the inflow of water to the caverns and may result in losses caused by dissolved gases dissolved in this water. In the absence of a low permeability rock mass, a tight storage medium can be constructed by cladding the cavern with a liner, which can be a polymer membrane or thin steel sheets [40,42–44]. The liner seals the gas whereas the surrounding rock absorbs the load. The host rock must be of sufficient quality as it determines the maximum storage

pressure and consequently the storage volume and cost of mining. Lined rock caverns have been successfully demonstrated in a pilot project in Scandinavia in early 2000 to store natural gas [45]; however, no commercial facility has been built to date.

This work analyzes the technical aspects and cost effectiveness of different methods for bulk storage of gaseous hydrogen. Figure 1 identifies the storage methods of interest and their relevance to different applications. Confining the scope of the work, we only consider the large-scale gaseous storage options suitable for outages of hydrogen production plants (10-30 days of daily capacity) and seasonal variations in hydrogen demand. We consider two geological storage options, salt caverns and lined rock caverns, and one geographically agnostic underground pipe storage method. Bottoms-up cost analyses are performed for each case and the results are discussed below and in the supplemental document.

## 2. Methodology

### *2.1 Bulk $H_2$ Storage in Underground Pipes*

Figure 2 presents a schematic of the core cost characteristics constructing the pipe facility.

The main cost components are a) pipe production at the mill, b) surface coating of the pipes, c) transmission of pipes by rail and truck to the staging area, d) site preparation (excavation and backfilling), and e) pipe installation, welding, weld inspection, and final pressure testing for integrity.

Starting with the production of the pipes, electric resistance welded (ERW) pipes were assumed in this analysis [46]. We also considered pipes with different diameters and wall thicknesses (18-36" O.D., schedule 10-100) as to determine the minimum cost. Steel plates are trimmed to proper dimensions, rolled into a pipe section and resistance welded. The pipe is cut in segments of 12 m, heated in a furnace and allowed to gradually cool as to temper the steel. Pipes are inspected and hydrostatically tested and packaged before delivery from the plant. Supplemental A details the cost

of the pipes for different diameters and schedules investigated in this study. The cost of the pipes, normalized in terms of mass, ranges between $1,090/MT to $1,371/MT. These estimates compare well with the costs of similar pipes (>24") used in the natural gas and oil industry (1,080-1644 $/MT) [47]. The variance is attributed to the cost of steel which accounts for ~64% of the cost of the pipe.

The pipes are packaged and transported from the production mill to the nearest coating plant for corrosion resistance. A three-layer polyethylene coating is sufficient for corrosion prevention, without the need to install more expensive cathodic protection systems. The coated pipes are transported by rail to and from the coating plant and locally by trucks to the storage site. For our base case scenario, a 24" pipe schedule 60 can store hydrogen at a maximum pressure of 100 bar, and ~24,000 pipe segments are required for storing 500 tonnes of hydrogen. This number of pipes may require purchase orders from several mills and a broker to facilitate the procurement, inspection and other logistics. The cost contribution of coating and logistics [47] are summarized in Table 1.

Site preparation costs include expenses for land acquisition, site clearing, grubbing and disposal of debris and excavation and backfilling of the pipe network. The pipes are covered by 1.2 m of soil, and pipe strings are spaced 80 cm apart as necessary clearance for welding. A land area of 110 acres is required for our base case storage facility, and a total volume of 1,055,040 cubic yard (cy) of soil is excavated. Land clearing costs were estimated from guides for road construction (USDA forest service) [48]. The cost factors used are based on the use of the hydraulic excavator for clearing/pioneering. The land was assumed to primarily consist of tops and limbs (40%), logs (35%) and stumps (25%), which resulted in a total volume of 30 CCF/acre of classification.[1] The

[1] Timber volumes are generally expressed in CCF (100 cubic feet).

total cost of the right of way for clearance was estimated at a base rate of $2,830/acre. A topographical factor of 10% was added to the base rate (less than 20% slope) and an additional 65% for disposal and clearing of the debris. Costs for site excavation, hauling of soil and backfilling (after pipes have been installed) were determined from the cost of equipment, labor rate and machinery efficiency and diesel fuel costs. In addition, a land cost of $50,000/acre was assumed; 25% of the site cost was attributed to engineering costs, logistics and project contingency. A total levelized cost for the site was estimated to $29.83/cy. The costs and assumptions for site preparation are detailed in Supplemental A.

Pipes are installed at the site before backfilling. Pipe installation costs are associated with the welding of individual pipes segments (12-m long) and end domes, radiographic weld inspection and hydraulic pressure testing for all pipe strings. Welding all the segments of pipelines (process efficiency, and cost) is adopted after the work by Fekete et al. [49]. A total welding cost of $1,192/joint was determined, 93% of which was attributed to labor. The methodology and cost of pipe installation is detailed in Supplemental A.

After backfilling, aboveground facilities are included in the site preparation costs, which consist of compressor, valves and pipes connecting the storage facility to an adjacent hydrogen production facility. The inlet suction pressure from the hydrogen plant is at 20 bar, and $H_2$ is compressed to 100 bar with two 802 kW compressors. Aboveground capital costs and operational expenses were estimated by the hydrogen delivery scenario model [50]. Table 2 gives the capital, operational and maintenance costs for the aboveground facility.

### *2.2 Bulk $H_2$ Storage in Underground Lined Rock Caverns (LRC)*

Lined rock caverns have as to date not been used for storage in industrial scale, however, pilot projects have confirmed its viability for NG storage with sufficient construction details [45,51,52]. Figure 3a shows the principle cost features of this type of storage system and consists of four parts: excavation of the underground facility including access tunnels and the cavern enclosure, installation of the liner and concrete surrounding the liner and the cavern forming the storage facility and the aboveground facilities including compressor, pipes and valves.

The excavation of the rock cavern consists of one or multiple cavern systems with an arrangement of shafts and tunnels joining them with the ground surface. The excavation is performed by conventional drill-and-blast operations [52,53]. The cross-sections of the shafts and tunnels are 25 $m^2$. The total length of the tunnels is 760 m (a 400 m access tunnel, two 80 m tunnels connecting to the lower and upper part of the dome and a 200 m shaft). The tunneling operations also include widenings periodically as to facilitate the turning of mining vehicles. This adds another 20% to the total length of the tunnels. Rock support is used in sections of the tunnels (30%) with grouted rebar. Grouting is also performed to minimize groundwater leaking and allows for more efficient pumping equipment. The caverns are excavated as hemispherical cylinders and reinforced with rock bolting and concrete. The dimensions of a cavern capable of storing 500 tonne of $H_2$ at a pressure of 150 bar are 62 m in diameter and 71 m in height. The excavated rock is assumed to have no monetary value, and the rocks are trucked 10 miles to remote deposit sites. Cost contributions of mining the tunnels and cavern with additional work to fortify the walls are detailed in Supplemental B.

Figure 3b shows some important technical details of the construction and structural properties for LRC storage. The lining represents a key component to ensure a proper gas storage and structural integrity. The steel liner contains the gas whereas the sealing layers transmits the

pressure load to the surrounding rock [42,54,55]. The lining consists of 15-mm thick mild steel sheets which are welded together to form the inner vessel of the dome. The concrete layer (2-m thick), reinforced with a rebar lattice, distributes the pressure load to the surrounding rock. A drainage arrangement of perforated pipes is also mounted around the cavern before the concrete is poured as to remove water accumulating in the lining section. The maximum pressure of the gas that can be stored depends on the cavern depth and the quality of the rock mass [42]. A depth of 150 m can typically a withstand pressure of 300 bar and higher preventing the rock mass from uplifting [52]. The surrounding rock mass must be strong enough as to not be deformed by the pressure load and cause fracture of the lining. This is determined by the rock mass rating (RMR) and thus a geological survey is imperative to locate the best site for the cavern [56,57]. The last panel in Figure 3b correlates the RMR method with the maximum cavern storage pressure. For our base case scenario, we consider that the cavern is located 150 m below the surface and assume an RMR value to the rock of 50, which limits the maximum storage pressure to 150 bar. The costs associated in commissioning the cavern to a fully functional storage system are detailed in Supplemental B.

Finally, the above-bound facility consists of a compressor and balance of plant components (e.g. piping and valves) similar to that for pipe storage systems. We assume that the hydrogen production facility is located one mile from the storage site and a pipeline is constructed with right-of-way associated costs [58]. Table 3 presents the capital, operational and maintenance costs for the aboveground facility.

### *2.3 Bulk $H_2$ Storage in Underground Salt Caverns*

Salt caverns are especially suitable for low-cost hydrogen storage in extensive quantities, but are geographically limited to a few locations; see Figure 4. Favorable locations in the U.S. are found in the Gulf Coast and the northeast.

Figure 5 shows the main cost contributions considered in the development of the salt cavern storage. Costs are associated with leaching out the cavern with fresh water at sufficient depth and size, brine transportation and disposal costs and the surface-bound facility.

Before the salt cavern is constructed, a geological survey is necessary to ensure that the mining will not threaten the integrity of the containing salt. Three-dimensional mapping and seismic surveying are common geophysical methods. An exploratory drilling is finally conducted to explore porosity and permeability, stress measurements and the presence of corrosive substances and the adherence of the salt dome [9,59].

Meeting all requirements, the underground cavern can be created using a process known as solution mining or leaching. Solution mining technology converts a simple borehole into a storage cavern through the injection and extraction of water. First, a borehole from the surface to the final required depth is drilled through the salt formation. This process is very similar to the established oil and gas drilling practice [9,60,61]. After drilling, an annular pipe is fitted through the borehole and lowered to the middle of the cavity. To form the salt cavern, fresh water is pumped through the inner tubing to the bottom of the borehole. The water dissolves the salt and saturated brine is removed through the other pipe, leaving an area of free space. Water supply for the leaching process is assumed to be available from a nearby well (2 miles), and pipelines are required to supply the water during the leaching operation. Approximately 7-8 $m^3$ of fresh water is needed to dissolve 1 $m^3$ of salt [62]. Brine is stored on site for up to 2 days of operation and is disposed by trucks into deep Class II wells [63,64]. It is possible to recover salt from the brine as a chemical

feedstock (e.g., for the chlor-alkali process); however, the facilities need to be in close proximity to the storage site for this to be economically viable. In absence of Class II wells, a more costly disposal would be trucking the brine to the nearest wastewater treatment plant [65].

Upon completion of the cavern structure, preparation is required to convert from brine to gas storage. Before any gas is introduced, an initial mechanical integrity test (MIT) is performed to confirm gas tightness of the uppermost salt and the well completion. The last step in the construction of the cavern is the debrining process, which essentially uses gas pressure to displace the brine contained in the cavern at the completion of solution mining. Caverns constructed in salt domes are typically 60 to 240 m in length, and there is a great deal of flexibility in positioning a cavern depth down to about 1,800 m. The storage pressure increases by 0.156 bar per meter of depth ( [60]. For our base scenario, the cavern was placed 765 meters belowground, requiring a gas pressure of 120 bar to prevent creep of the cavern walls. The cost to construct a salt cavern is detailed in Supplemental C.

The surface-bound facility is identical to the LRC and pipe storage facilities, as discussed previously. The exception is that a water-wash and gas-drying step is needed for hydrogen purity control. The cavern contains a heel of brine in the sump (bottom layer of debris), which cannot be removed during the debrining process, therefore gas withdrawn from the cavern will be saturated with salt during an initial period of operation. The water-wash column removes entrained salt, and the dryer (sorbents) removes water in order to meet $H_2$ quality specifications. This adds an operational cost of $44,783/year in desiccant drying costs.

### *2.4 Life-Cycle Costs*

For consistent, transparent and comparable financial results across all options, storage life-cycle costs were evaluated using the U.S. Department of Energy (DOE) standard economic tool (H2A)

and assumptions. H2A is a financial spreadsheet evaluating the net present value of a capital investment and is freely available for download at DOE's website [66]. The following set of key economic parameters was selected for our analyses: a) reference year: 2019, b) after-tax internal rate of return: 10%, c) inflation rate: 1.9%, d) economic analysis period: 30 years, e) 15 year depreciation schedule (MARCS)[2] with an effective total tax rate of 38.9% and f) electricity cost: $0.0574/kWh. Indirect costs are also included as fraction of the total installed underground cost. We have used the default H2A financial assumptions of 15% for engineering, 10% for contingency, and 3% for permitting.

## Results and Discussion

### *3.1 Bulk $H_2$ Storage in Underground Pipes*

We investigated the cost of an underground pipe facility capable of storing 500 t-$H_2$. We considered API 5L Grade X52 pipes (52 ksi minimum yield strength, 66 ksi tensile strength) available in 18-36" outer diameter (OD) and Schedules 20, 40, 60, 80 and 100. We followed the ASME B31.12 code for hydrogen piping and pipelines to determine the maximum permissible pressure considering the nominal pipe OD, nominal wall thickness, longitudinal joint factor, design factor, material performance factor and minimum yield strength. Table 4 lists the maximum permissible storage pressures after a 12.5% allowance in nominal wall thickness.

Figure 6 presents the capital cost of the underground storage system using 24"-OD pipes of different schedules. It shows that the capital cost is lowest, $516/kg-$H_2$ stored, for a storage facility with 24"-OD, Schedule 60 pipes. This system can be configured with 300 parallel pipe strings, each string 3,150 ft in length for a total of 180 miles equivalent length, requiring 110 acres of land.

[2] *The modified accelerated cost recovery system (MACRS) is a depreciation system used for assessing taxes in the U.S. Federal and state taxes are based on year 2017 schedules.*

The majority the cost, 63%, is due to delivered pipe cost factors: 50% for pipe material and milling, 10% for coating and <3% for shipping. Pipe installation accounts for 23% of the total cost: 11% for welding, 10% hydrostatic pressure testing and >1% for weld inspection and joint coating. Site preparation represents 14% of the overall cost: 8% for site excavation and backfilling, 4% for land and projects and 2% for aboveground facilities.

Because the majority of the capital cost is due to manufacturing of pipes, followed by welding, pressure testing, site excavation and backfilling, minimizing the number of pipes and/or pipe mass is important in reducing costs. For example, choosing a Schedule 20 rather than Schedule 60 pipe increases the capital cost by 59%: 26% due to higher delivered pipe cost, 13% due to higher pipe installation cost and 20% due to higher site preparation cost. Similarly, choosing a Schedule 100 rather than Schedule 60 pipe slightly increases the capital cost by 13%: 8% due to higher delivered pipe cost (more pipe mass), 8% due to higher pipe installation cost (more expensive welding), offset by 3% smaller site preparation cost (fewer pipes, smaller land area).

Figure 6 includes the levelized the cost of underground pipe storage on a 50 tonnes per day (tpd)basis. This levelized cost is \$2.17/kg-$H_2$ for Schedule-60 pipes, nearly 95% of which is due to capital expenditure CAPEX levelized assuming a 10% annual rate of return. Because the pipe material cost is a major contributor, we did a single-variable study using the historical swings in line prices: \$1127/ton, \$803/ton and \$1,360/ton for current, historical low, and historical high prices, respectively. These historical swings cause the $H_2$ cost to decrease by \$0.30/kg or increase by \$0.22/kg. Clearly, the underground pipe storage method is unlikely to find wide deployment considering that the target cost of dispensed $H_2$ is \$4/kg.

Table 5 summarizes the detailed breakdown of performance (storage pressures, total equivalent length) and cost parameters for storage systems with different pipe ODs and schedules that lead to

least capital investment. In general, the cost parameters vary with ODs in a non-linear fashion, but the storage system with 24" OD, Schedule 60 pipes is an optimal combination of shortest length and least pipe material leading to the smallest costs for milling, coating, weld inspection and hydrostatic testing and site preparation.

### *3.2 Bulk $H_2$ Storage in Underground Lined Rock Caverns*

We conducted a study to estimate the cost of storing 500 t-$H_2$ in an underground LRC facility varying the maximum storage pressure while holding the minimum pressure as constant at 20 bar. The results of the study are summarized in Figure 7 and Table 6. Figure 7 shows that the capital cost decreases at higher storage pressures; however, the rate of decrease slows at pressures above 150-200 bar.

At 150-bar storage pressure, the underground facility accounts for 66% of the total capital cost including 34% for cavern excavation, 16% for tunnels, 8% for liner and 8% for concrete. The aboveground facility represents 14% of the total cost including 11% for compressor, 3% for piping and instrumentation and <1% for cushion gas. The miscellaneous items represent 21% of the total cost including 10% for engineering, 6% for contingency, 3% for survey and 2% for permitting. Thus, cavern excavation, tunnels, compressor and engineering are the biggest cost items.

Lowering the storage pressure from 150 bar to 75 bar results in 61% higher capital cost, 82% of which is due to the higher cost of the underground facility which has more than double the storage volume. The aboveground facility cost is ~3% lower primarily because of a smaller compressor bust; this cost saving is offset by a 14% increase in the miscellaneous costs for engineering services and contingency.

The capital cost is smallest at 300-bar storage pressure. Compared to the 150-bar storage system, the required storage volume is about 50% smaller resulting in a 21% cost saving for the

underground facility (cavern excavation, liner and concrete), and the 5% higher cost for the aboveground facility (compressor and piping) is nearly offset by 6% saving in the miscellaneous costs for engineering and contingency.

Table 6 provides a detailed breakdown of the major cost components, grouped as underground facility, aboveground facility, and miscellaneous, for different storage pressures. For the purpose of this study, the costs of the underground tunnel, survey and land do not vary with storage pressure.

Figure 7 includes the levelized the cost of underground LRC storage on a 50 tpd basis. The levelized cost is \$0.36/kg-$H_2$ for 1a 50-bar storage pressure, nearly 85% of which is due to the capital expenditures levelized assuming 10% annual rate of return. Within the parameters of this study, the levelized cost is \$0.0.05/kg-$H_2$ smaller for a 100-bar storage pressure and \$0.0.05/kg-$H_2$ larger for a 250-bar storage pressure. Compared to underground pipe storage, the capital cost of underground LRC is 90% smaller and the levelized cost is 85% smaller. However, finding a suitable rock mass strength that is able to contain the hydrogen gas at sufficiently high pressures, may be restricted and limited geographically.

### *3.3 Bulk $H_2$ Storage in Underground Salt Caverns*

We conducted a study to estimate the cost of storing 500 t-$H_2$ in underground salt caverns at depths between 500 to 1,200 m. The results of the study are summarized in Figure 8 and Table 7 as a function of the storage pressure that depends on the cavern depth. The study treats 30% of the storage capacity as minimum cushion gas. Figure 8 shows a general trend of the underground and aboveground facilities cost increasing, and the brine disposal cost decreasing, with higher pressure. The net result is that the capital cost varies narrowly between \$35/kg-$H_2$ and \$38/kg-$H_2$ with a shallow minimum at a 120-bar storage pressure.

At a 120-bar storage pressure, the underground facility accounts for 50% of the total capital cost including 16% for drill and casing, 13% for geological survey and mechanical integrity testing, 12% for brine leaching and 9% for engineering and permitting. The aboveground facility represents 27% of the total capital cost including 18% for compressor, 5% for piping and drying and 4% for cushion gas. The brine-related activities represent 25% of the total capital cost including 12% for transportation, 12% for disposal and 1% for brine storage. Thus, the major cost factors are related to brine leaching, handling (transportation) and disposal followed by compressor, and salt cavity formation (drill and casing, integrity testing and engineering).

Table 7 provides a detailed breakdown of the major cost components grouped as underground facility construction, engineering, brine handling and aboveground facility. Higher storage pressures require smaller cavern volumes and incur lower costs for brine leaching, transportation and disposal. However, higher storage pressures also require larger compressors and deeper caverns that lead to additional costs for bore and production tubing.

Figure 8 includes the levelized the cost of underground salt cavern storage on a 50 tpd-$H_2$ basis. The levelized cost is \$0.21/kg-$H_2$ for a 120-bar storage pressure, nearly 75% of which is due to the capital expenditures levelized assuming a 10% annual rate of return. Within the parameters of this study, the levelized cost is \$0.0.02/kg-$H_2$ smaller if the cost for brine disposal can be avoided because of an existing market demand, and it is \$0.21/kg-$H_2$ higher if the brine disposal cost is \$2/bbl rather than the default value of \$0.50/bbl. Brine disposition as affected by local political conditions and environmental concerns introduce a financial uncertainty in any project using salt caverns.

Comparing the results in Figures 6-8, the capital cost of the underground salt cavern option is 93% smaller than underground pipe storage and 51% smaller than underground LRC storage.

Similarly, the levelized cost of the underground salt cavern option is 90% smaller than underground pipe storage and 40% smaller than underground LRC storage. However, like the LRC storage option, the applicability of underground salt caverns is geographically restricted, and may require a detailed environmental impact study.

### *3.4 Bulk $H_2$ Storage: Outlook*

We conducted a study to investigate the effect of economy of scale on the cost of the three storage options analyzed using working storage pressures of 100-8 bar for underground pipes, 200-30 bar for LRC and 120–28 bar for salt caverns. A lined rock cavern is situated 150-m deep and can hold up to 600-t $H_2$; a salt cavern is located at a 800-m depth and can hold up to 1,000-t $H_2$. Two or more caverns are placed in service if the required storage capacity exceeds the maximum capacity of a single cavern. The results of the study are summarized in Figure 9 in terms of the installed capital cost and annual storage cost (\$/kg-$H_2$ stored) over a 30-year lifetime of the storage facilities. The storage cost includes the levelized capital cost assuming a 10% annual rate of return and recurring operating (labor and utilities) and maintenance costs and can be calculated from the following correlation using the coefficients in Table 8.

$$Annual\ Cost = \exp\left(a(\ln(m))^2 - bln(m) + c\right)$$

Figure 9 shows only a small effect of the economy of scale on the installed capital and annual storage costs of the underground pipe storage option, i.e., a less than 10% cost reduction in scaling usable $H_2$ from 1 tonne to 20 tonnes. As discussed earlier, fabrication of coated pipes is a major cost factor that does not benefit much from manufacturing at high volumes. As indicated in Figure 8, underground pipes can be more economical than geological facilities for usable $H_2$ storage amounts less than 20 tonnes. However, the capital (~\$600/kg-$H_2$ stored) and annual (~\$90/kg-$H_2$ stored) costs are far too high for underground pipe storage to be a viable storage solution for large

quantities of hydrogen. This option could be more applicable when storing hydrogen at smaller scales, for instance to meet daily peak demands.

Lined rock and salt caverns show substantial reduction in cost as the storage capacity is increased. The salt caverns are more cost-effective than lined rock caverns, although the two options are never available at the same location. Also, salt caverns scale better because of their higher maximum $H_2$ storage capacity, i.e., salt caverns need to be added at a slower rate than lined rock caverns as the storage capacity is increased.

## 3. Conclusions

Massive deployment of hydrogen and fuel cell technologies will require viable methods to store hydrogen at different time and length scales. We have analyzed the technical aspects and economics of bulk hydrogen storage in underground pipes, lined rock caverns and salt caverns in different quantities, 1-3,000 tonnes, equivalent to <1 to 60 days of storage for a 50 tonnes per day production facility. The main conclusions are summarized below.

- Hydrogen storage in underground Type-1, API 5L X52 pipes is more economical than in geological caverns for usable amounts <20-t-$H_2$. At ~\$560/kg-$H_2$ stored, the estimated capital cost is high as is the annual cost at ~\$84/kg-$H_2$ stored. Because the pipe material is a major cost factor, the capital and operating costs for this storage method do not decrease appreciably with an increase in the amount of stored $H_2$.
- Unlike underground pipes, the installed capital cost of salt caverns decreases appreciably from ~\$95/kg-$H_2$ at 100 t-$H_2$ stored to <\$19/kg-$H_2$ at 3,000 t-$H_2$ stored. Over the same scale, the annual storage cost decreases from ~\$17/kg-$H_2$ at 100 t-$H_2$ stored to ~\$3/kg-$H_2$ at 3,000 t-$H_2$ stored.

- Like salt caverns, the installed capital cost of lined rock caverns decreases from ~\$160/kg-$H_2$ at 100 t-$H_2$ stored to <\$44/kg-$H_2$ at 3,000 t-$H_2$ stored. Over the same scale, the annual storage cost decreases from ~\$26/kg-$H_2$ at 100 t-$H_2$ stored to ~\$7/kg-$H_2$ at 3,000 t-$H_2$ stored.
- Storing >750-t usable $H_2$ requires multiple caverns. Hydrogen storage in salt caverns is more economical than in lined rock caverns. Also, the cost of salt caverns scales more favorably with size because the salt caverns are larger than lined rock caverns and need to be added at a slower rate as the storage capacity is increased.

## 4. Acknowledgments

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## 5. Attributions

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| Unit/Description | Value | Unit |
|---|---|---|
| Pipe broker (procurement, inspection, logistics) | 10 | % of pipe cost |
| Packing for shipping (mill) | 30 | $/ton |
| Rail shipping | 0.05 | $/ton-mile |
| Truck shipping | 0.12 | $/ton-mile |
| Surface coating plant distance | 500 | miles |
| Factory surface coating | 45 | $/m$^2$ |
| Repacking at coating plant | 15 | $/ton |
| Rail distance from mill to coating plant | 500 | miles |
| Rail distance from coating plant to staging area | 400 | miles |
| Truck distance from staging area to site | 100 | miles |

Table 1. Coating and logistic cost contribution for pipes

| Unit/Description | Value | Unit |
|---|---|---|
| Storage capacity | 500,000 | kg |
| Cushion gas needed | 46,613 | kg |
| Compressor cost (installed) | 3,249,872 | $ |
| Pipes, valves and metering | 1,079,173 | $ |
| Cushion gas | 233,064 | $ |
| *Aboveground costs* | *4,562,109* | $ |
| Labor cost | 246,769 | $/year |
| Maintenance cost | 173,162 | $/year |
| Electricity cost | 11,038 | $/year |
| Property tax | 55,074 | $/year |
| Insurance | 91,242 | $/year |
| *Total O&M* | *577,285* | *$/year* |

Table 2. Aboveground capital and O&M costs for underground pipe storage. Cost of hydrogen cushion gas evaluated at $5/kg, labor cost of 33.5 h and electricity at 5.74 cents/kWh.

| Unit/Description | Value | Unit |
|---|---|---|
| Storage capacity | 500,000 | kg |
| Cushion gas needed | 84,085 | kg |
| Compressor cost (installed) | 3,535,794 | $ |
| Pipe to $H_2$ production facility | 749,130 | $ |
| Valves and instrumentation | 112,370 | $ |
| Cushion gas | 420,423 | $ |
| Land cost | 313,769 | $ |
| *Aboveground costs* | *5,131,486* | *$* |
| Labor cost | 252,000 | $/year |
| Maintenance cost | 150,047 | $/year |
| Electricity cost | 11,982 | $/year |
| Property tax $ insurance | 102,630 | $/year |
| *Total O&M* | *516,659* | *$/year* |

Table 3. Aboveground capital and O&M costs for LRC storage. Cost of hydrogen cushion gas evaluated at $5/kg, labor cost of 33.5 h and electricity at 5.74 cents/kWh.

| | Pipe Outer Diameter | | | |
|---|---|---|---|---|
| **Schedule** | **18"** | **24"** | **32"** | **36"** |
| 20 | 43 | 39 | 39 | 34 |
| 40 | 77 | 71 | 53 | 47 |
| 60 | 103 | 100 | - | - |
| 80 | 117 | 114 | - | - |
| 100 | 140 | 139 | - | - |

Table 4. Available pipe sizes and maximum allowable pressure (atm)

| **Pipe Diameter (inches)** | **18** | **24** | **32** | **36** |
|---|---|---|---|---|
| Schedule | 60 | 60 | 40 | 40 |
| Maximum storage pressure (atm) | 103 | 100 | 53 | 47 |
| Mill cost ($/kg) | 275 | 259 | 288 | 294 |
| Coating ($/kg) | 67 | 53 | 71 | 71 |
| Shipping ($/kg) | 12 | 12 | 11 | 12 |
| Welding ($/kg) | 59 | 67 | 51 | 52 |
| Weld inspection and hydrostatic testing ($/kg) | 92 | 54 | 55 | 49 |
| Site preparation ($/kg) | 88 | 62 | 79 | 79 |
| Aboveground facilities ($/kg) | 9 | 9 | 9 | 9 |
| *Total Cost ($/kg)* | *602* | *516* | *564* | *566* |

Table 5. Breakdown of capital costs ($/kg-$H_2$ stored) for underground pipe storage

| **Maximum Storage Pressure (atm)** | **75** | **100** | **150** | **200** | **250** | **300** |
|---|---|---|---|---|---|---|
| Dome water volume ($m^3$) | 116,741 | 81,458 | 51,630 | 38,396 | 30,925 | 26,130 |
| Concrete volume ($m^3$) | 26,151 | 20,724 | 15,454 | 12,783 | 11,134 | 10,001 |
| Shaft ($/kg) | $11.2 | $11.2 | $11.2 | $11.2 | $11.2 | $11.2 |
| Dome ($/kg) | $54.0 | $38.4 | $25.0 | $19.0 | $15.6 | $13.4 |
| Liner ($/kg) | $9.1 | $7.2 | $5.4 | $4.6 | $3.9 | $3.6 |
| Concrete ($/kg) | $8.5 | $6.7 | $5.0 | $4.2 | $3.6 | $3.3 |
| *Total underground costs ($/kg)* | *$82.8* | *$63.5* | *$46.6* | *$38.9* | *$34.3* | *$31.4* |
| Engineering ($/kg) | $12.4 | $9.5 | $7.0 | $5.8 | $5.1 | $4.7 |
| Contingency ($/kg) | $8.3 | $6.4 | $4.7 | $3.9 | $3.4 | $3.1 |
| Permitting ($/kg) | $2.5 | $1.9 | $1.4 | $1.2 | $1.0 | $0.9 |
| Geological survey ($/kg) | $2.3 | $2.3 | $2.3 | $2.3 | $2.3 | $2.3 |
| *Total other costs ($/kg)* | *$25.5* | *$20.1* | *$15.4* | *$13.2* | *$11.9* | *$11.1* |
| Compressor ($/kg) | $5.0 | $5.8 | $7.1 | $7.8 | $8.5 | $9.2 |
| Piping, instrumentation, valves ($/kg) | $0.9 | $1.2 | $1.7 | $2.2 | $2.7 | $3.1 |
| Cushion gas ($/kg) | $1.0 | $0.7 | $0.4 | $0.3 | $0.3 | $0.2 |
| Land cost ($/kg) | $0.7 | $0.7 | $0.6 | $0.6 | $0.6 | $0.6 |
| *Total aboveground costs ($/kg)* | *$7.6* | *$8.3* | *$9.8* | *$11.0* | *$12.1* | *$13.1* |
| *Total cost ($/kg)* | *$115.8* | *$91.9* | *$71.8* | *$63.1* | *$58.3* | *$55.5* |

Table 6. Breakdown of capital costs ($/kg-$H_2$ stored) for underground LRC

| **Cavern Roof Depth (ft)** | **1,500** | **2,000** | **2,500** | **3,000** | **3,500** | **4,000** |
|---|---|---|---|---|---|---|
| Maximum storage pressure (atm) | 70 | 100 | 120 | 140 | 170 | 190 |
| Minimum storage pressure (atm) | 22 | 29 | 35 | 41 | 47 | 53 |
| Cavern water volume ($m^3$) | 125,897 | 96,185 | 78,294 | 67,634 | 59,287 | 52,973 |
| Cushion gas (%) | 31.1 | 31.1 | 30.7 | 30.8 | 30.8 | 31.0 |
| Drill+casing ($/kg) | $3.3 | $4.4 | $5.6 | $6.7 | $7.8 | $8.9 |
| Leaching ($/kg) | $5.1 | $4.5 | $4.1 | $4.0 | $3.8 | $3.8 |
| Mechanical integrity test ($/kg) | $2.3 | $2.3 | $2.3 | $2.3 | $2.3 | $2.3 |
| *Total construction costs* | *$10.7* | *$11.2* | *$11.9* | *$12.9* | *$13.9* | *$15.0* |
| Geological survey ($/kg) | $2.3 | $2.3 | $2.3 | $2.3 | $2.3 | $2.3 |
| Engineering ($/kg) | $1.6 | $1.7 | $1.8 | $1.9 | $2.1 | $2.2 |
| Contingency ($/kg) | $1.1 | $1.1 | $1.2 | $1.3 | $1.4 | $1.5 |
| Permitting ($/kg) | $0.3 | $0.3 | $0.4 | $0.4 | $0.4 | $0.4 |
| *Total engineering costs ($/kg)* | *$5.3* | *$5.4* | *$5.6* | *$5.9* | *$6.2* | *$6.5* |
| Brine transportation ($/kg) | $7.0 | $5.3 | $4.3 | $3.7 | $3.3 | $2.9 |
| Brine disposal ($/kg) | $7.0 | $5.3 | $4.3 | $3.7 | $3.3 | $2.9 |
| *Total brine disposal costs ($/kg)* | *$13.9* | *$10.6* | *$8.7* | *$7.5* | *$6.6* | *$5.9* |
| Compressor ($/kg) | $5.0 | $5.7 | $6.3 | $6.9 | $7.4 | $7.8 |
| Piping and instrumentation ($/kg) | $1.0 | $1.0 | $1.0 | $1.0 | $1.0 | $1.0 |
| Cushion gas ($/kg) | $1.4 | $1.4 | $1.3 | $1.3 | $1.3 | $1.3 |
| Land cost ($/kg) | $0.8 | $0.7 | $0.7 | $0.7 | $0.7 | $0.6 |
| *Total aboveground costs ($/kg)* | *$8.1* | *$8.8* | *$9.4* | *$9.9* | *$10.4* | *$10.8* |
| *Total cost ($/kg)* | *$38.0* | *$36.0* | *$35.6* | *$36.2* | *$37.0* | *$38.1* |

Table 7. Breakdown of capital costs ($/kg-$H_2$ stored) for underground salt caverns

| Storage | a | b | c |
|---|---|---|---|
| Underground pipe storage | 0.001559 | 0.035313 | 4.5183 |
| Underground lined rock caverns | 0.092286 | 1.5565 | 8.4658 |
| Underground salt caverns | 0.085863 | 1.5574 | 8.1606 |

Table 8. Coefficients for annual cost ($/kg-$H_2$ stored) correlations: $m$ is tonnes of stored $H_2$

NG Spherical Pressure Vessel, Germany[p1]



NG Pipe Storage, Erdgas, Switzerland[p2]

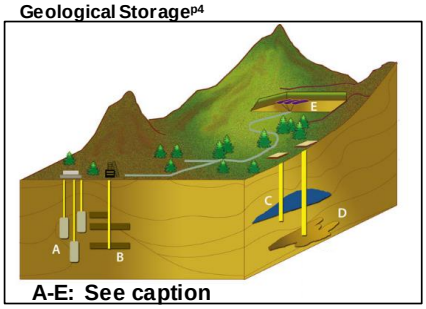


$LH_2$ Cryogenic Storage, NASA[p3]



| | Pressure Vessels | | | | Cryogenic | Geologic Storage | | |
|---|---|---|---|---|---|---|---|---|
| | Spherical Vessels | Pipe Storage | Pre-stressed Concrete | WireTough | $LH_2$ | Aquifer | Salt Cavern | Lined Rock Cavern |
| **Pressure, bar** | 1-10.4 | 7-100 | 7-875 | 7-875 | 20 | 150-170 | 55-152 | 10-230 |
| **Diameter, m** | 39.5 | 1.4 | 2.2 | 0.43 | 20 | | | 35 |
| **Wall Thickness, mm** | 34 | | 110 | | | | | 6-12 |
| **Length, m** | | 200 (13x15) | | 9.2 | | | | |
| **Depth, m** | | | | | | 1,500 | 1,200 | 115 |
| **Height, m** | | | 5.3 | | | | | 52 |
| **Water Volume, $m^3$** | 32,000 | 6,100 | 22 | 0.77 | 3400 | 4,141,000 | 566,000 | 40,000 |
| **Net Volume (STP), $m^3$** | 273,664 | 500,556 | 10,979 | 428 | 2,558,399 | 211,346,012 | 41,379,324 | 7,119,024 |
| **$H_2$ Stored, Tonnes** | 27 | 50 | 1 | 0.0389 | | 54,000 | 6,000 | 672 |
| **Working Capacity, Tonnes** | 25 | 45 | 0.987 | 0.0385 | 230 | 19,000 | 3,720 | 640 |
| **Application** | **City Gate** | **City Gate** | **Forecourt** | **Forecourt** | **City Gate** | **City Gate** | **City Gate** | **City Gate** |

Geological Storage[p4]




A-E: See caption

Lined Rock (Shown Tunnel Excavation)[p5]



**Figure 1**. Attributes of bulk storage methods: working pressures, storage volume, storage amount and working capacity. Geological storage: A) Salt caverns, B) Mines, C) Aquifers, D) Depleted mines, E) Hard rock caverns

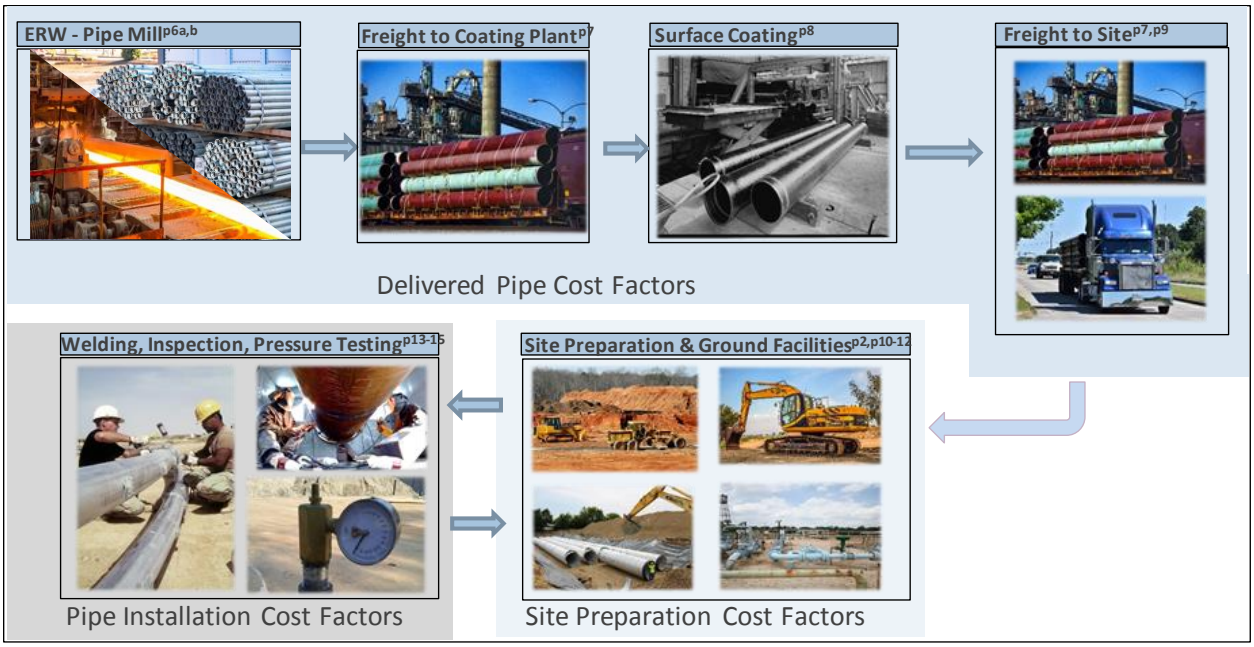



**Figure 2**. Schematic of cost components considered in the installation of underground pipe storage




**Figure 3a**. LRC main cost factors

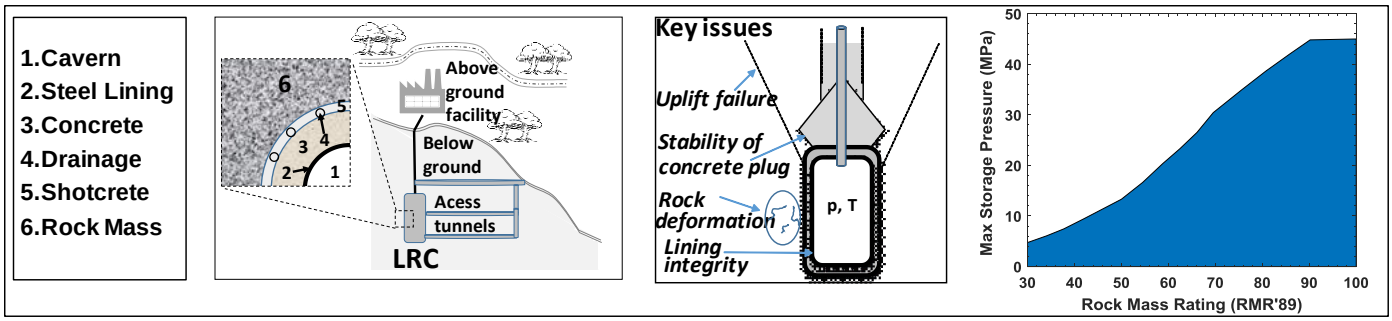



**Figure 3b**. LRC main technical factors

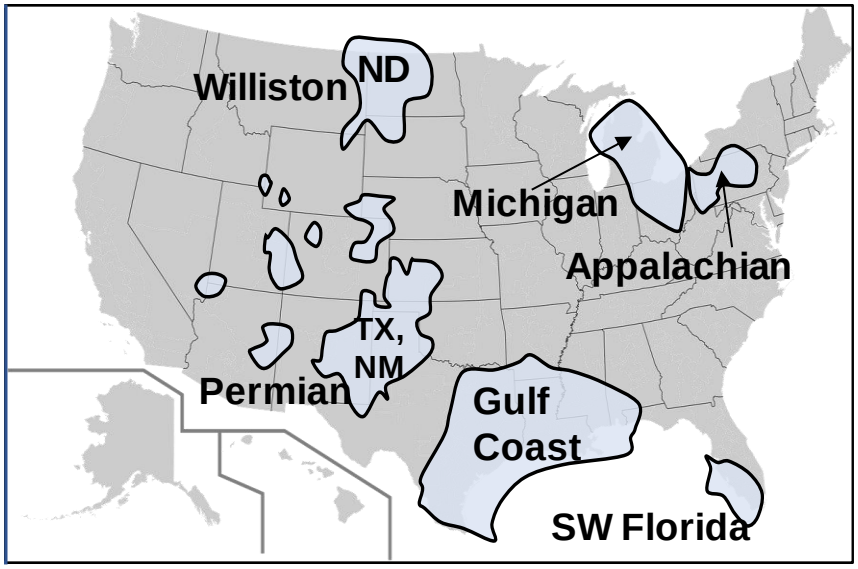



**Figure 4**. Location of major salt deposits in the United States. Redrawn after [22]. ND=North Dakota, TX=Texas, NM=New Mexico

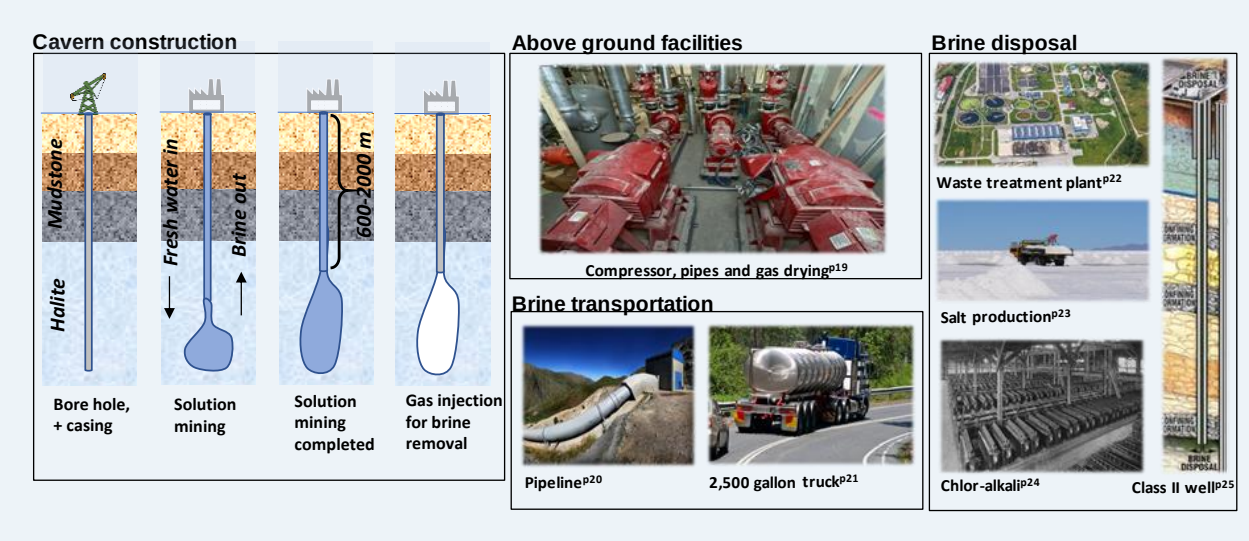
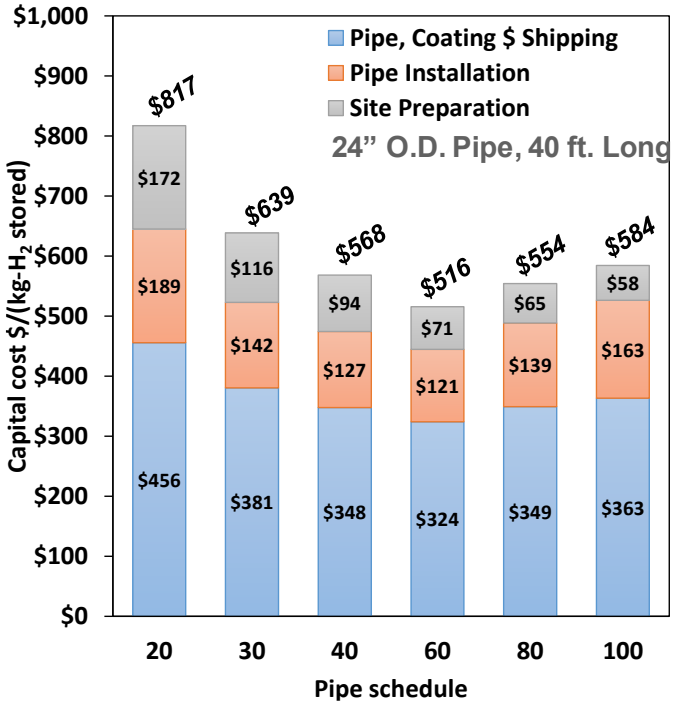
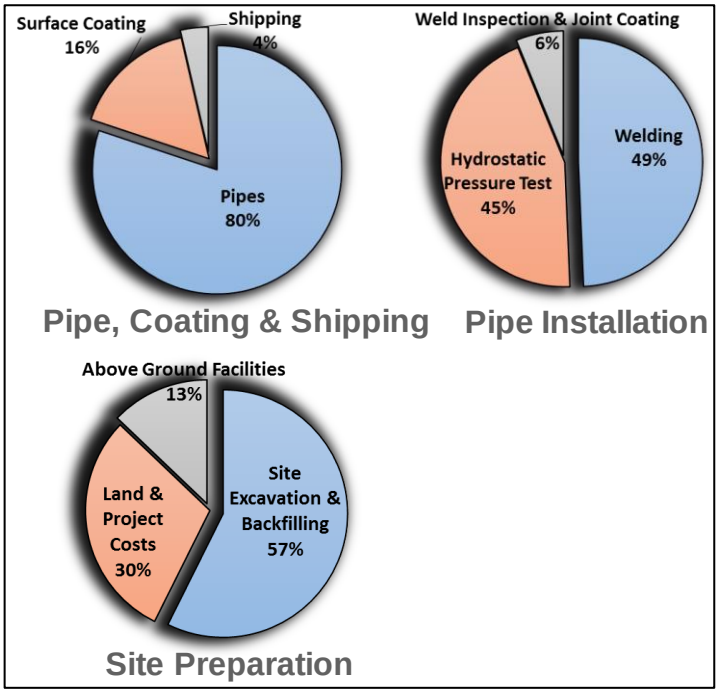
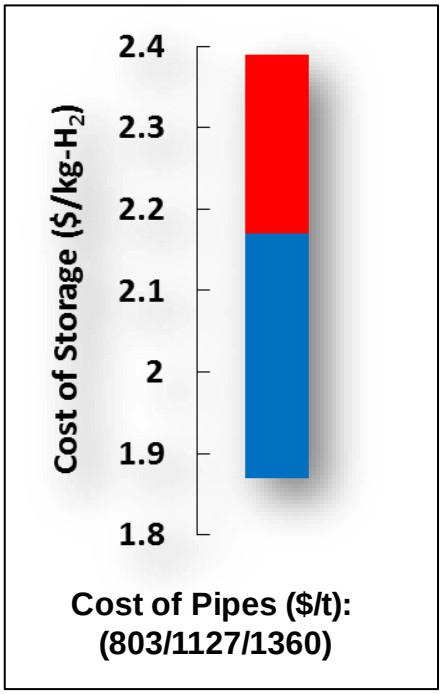



**Figure 5**. Underground salt cavern technical and main cost factors



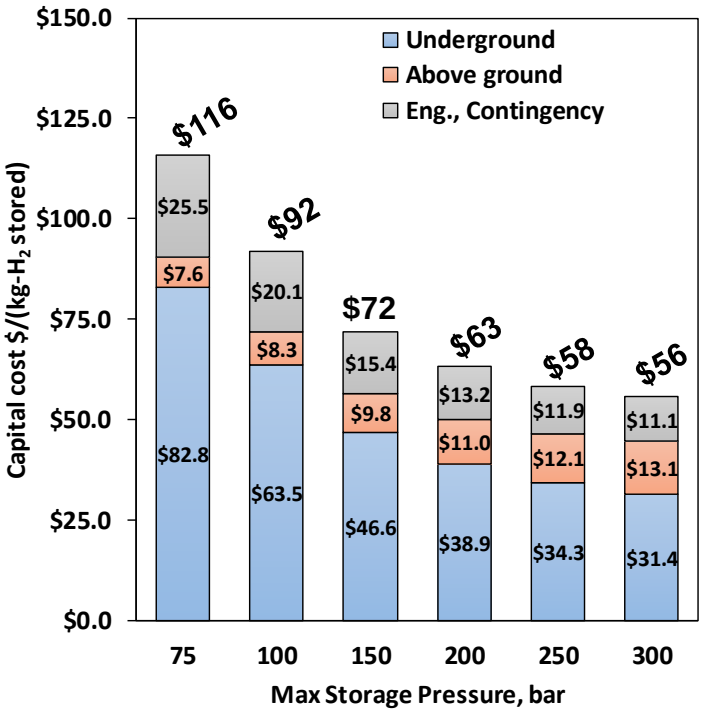

(a) Breakdown of main capital costs



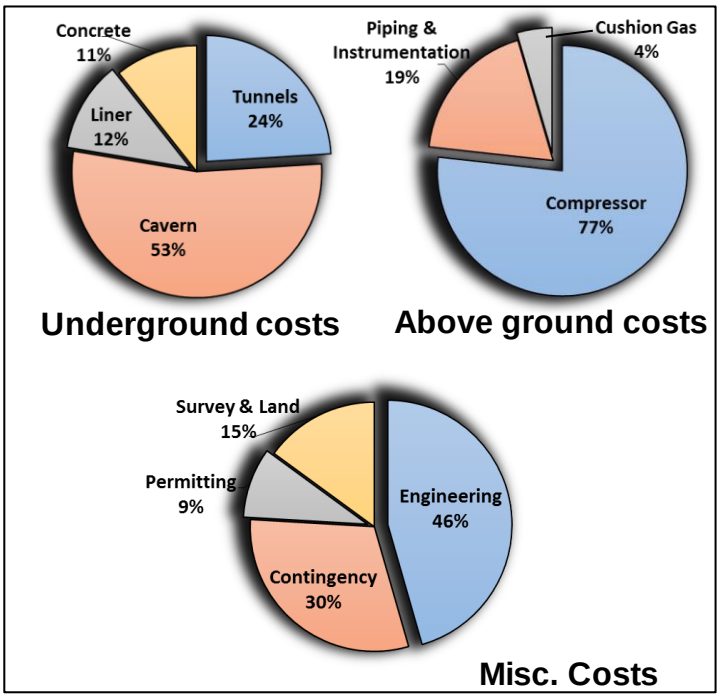




(b) Breakdown of capital costs for 24"-O.D., Schedule 60 pipe (c) Levelized cost of $H_2$

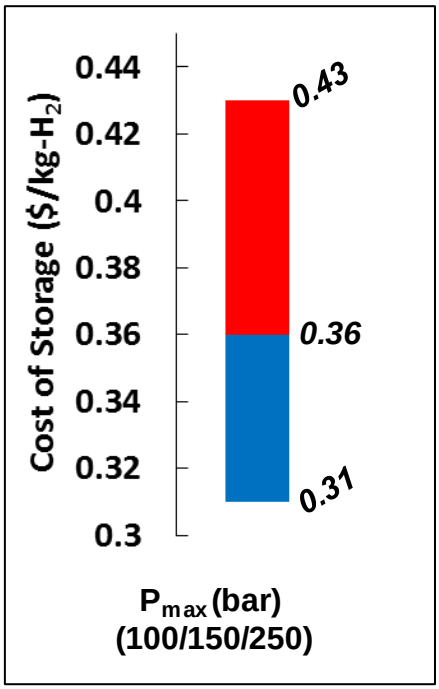
**Figure 6**. Cost of 500-t $H_2$ underground pipe storage facility




(a) Breakdown of main capital costs

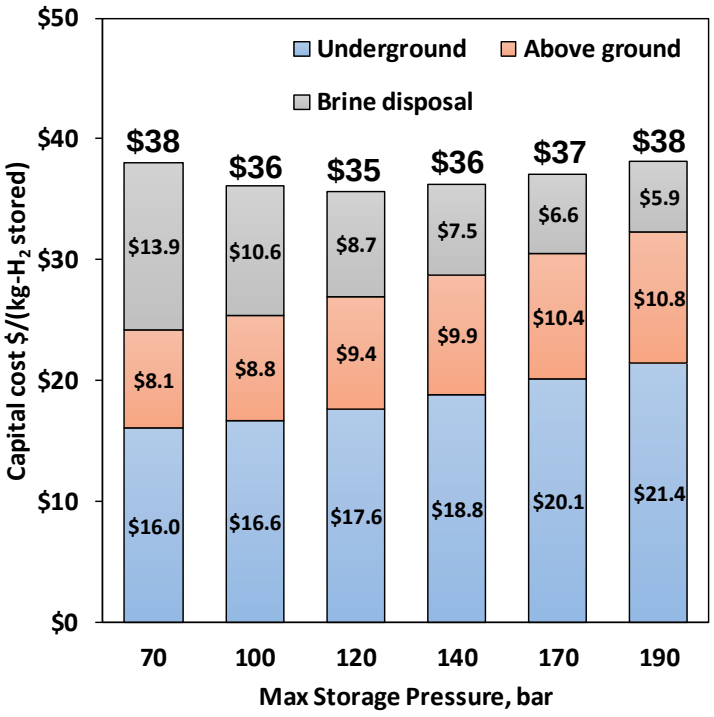



(b) Breakdown of capital costs; $P_{max}$ = 150 bar

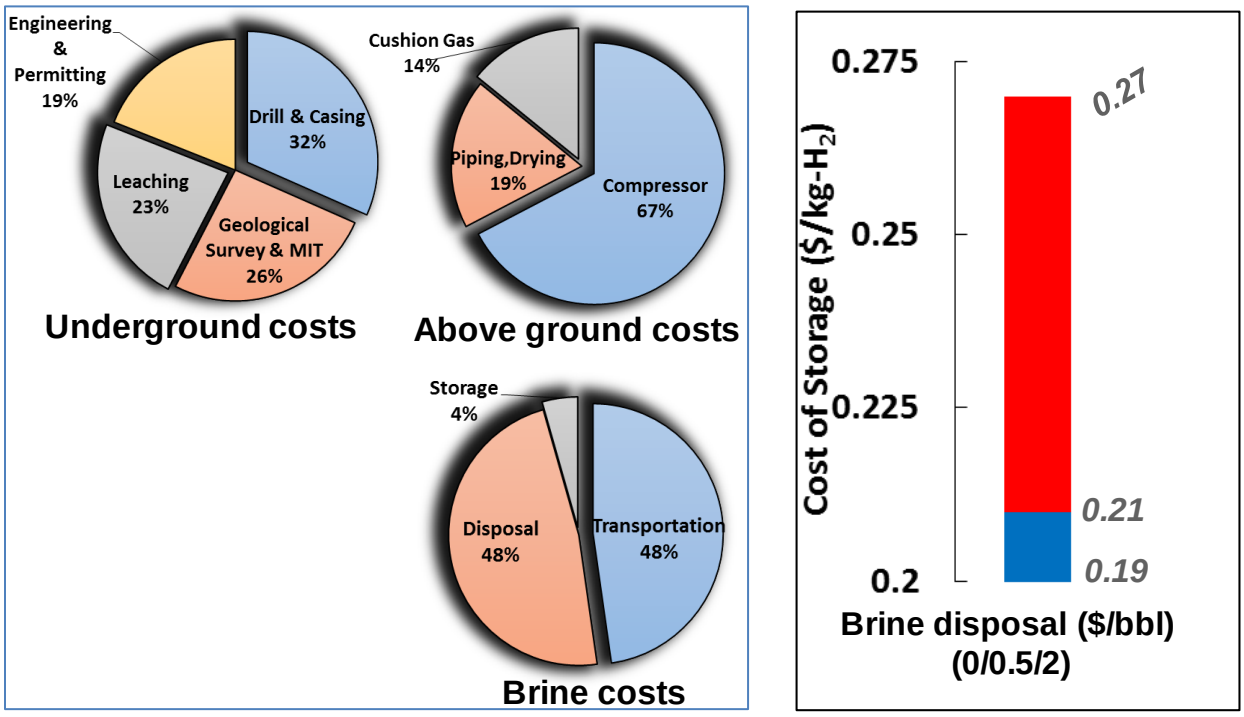



(c) Levelized cost of $H_2$

**Figure 7**. Cost of 500-t $H_2$ underground lined rock cavern facility



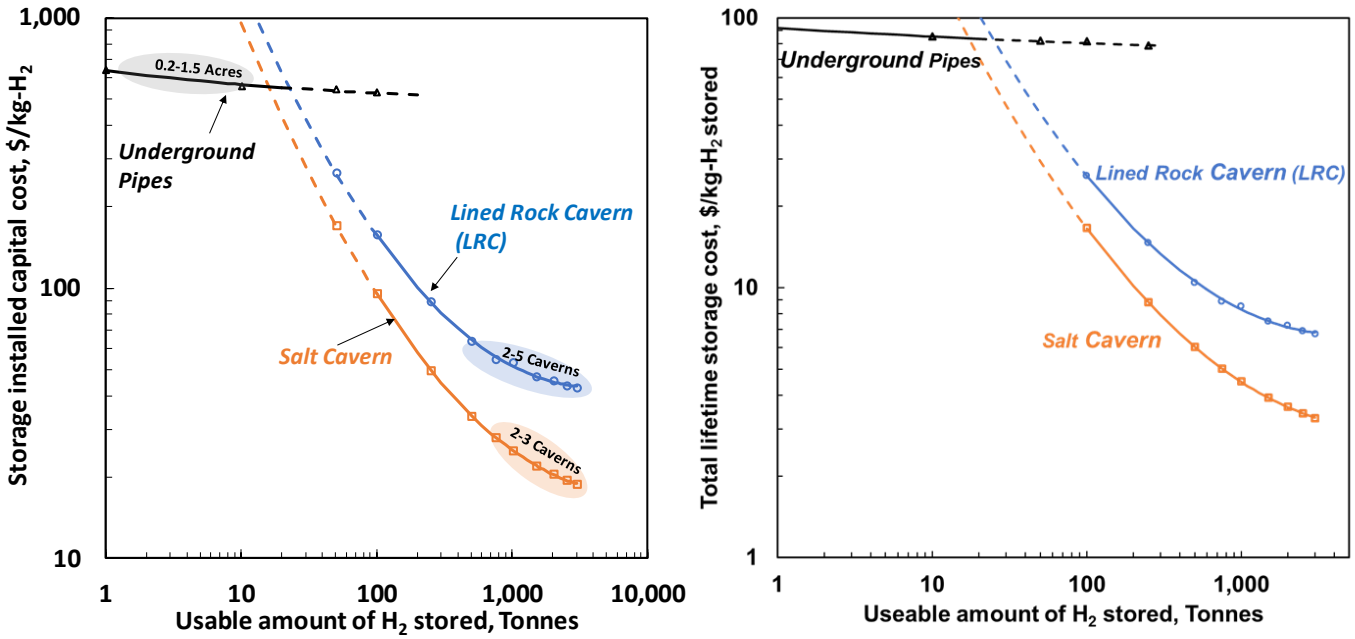

(a) Breakdown of main capital costs




(b) Breakdown of capital costs; $P_{max}$ = 150 bar

(c) Levelized cost of $H_2$

**Figure 8**. Cost of 500-t $H_2$ underground salt cavern facility




(a) Installed capital cost (b) Lifetime storage cost

**Figure 9**. Installed capital cost and lifetime storage cost: scaling with usable amount of stored $H_2$